

LESSON 6-6 • SESSION 2 • PRACTICE • FORM B

Testing and Justifying Equivalence

Learning target – I can test whether two expressions are equivalent by substituting values and by reasoning. • 6.EE.A.4

REMEMBER

- Equivalent expressions have the **same value for every** value of the variable.
- **Test** with the same value in both. One value with different results proves they are **not** equivalent.
- Matching values do not prove it. **Justify** with properties, like the Distributive Property.

WORKED EXAMPLE

Are $4(n + 3)$ and $4n + 12$ equivalent?**Step 1** Test $n = 2$: $4(2 + 3) = 20$ and $4 \times 2 + 12 = 20$.**Step 2** One match is not proof, so use a property.**Step 3** Distributive Property: $4(n + 3) = 4n + 4 \times 3 = 4n + 12$.**Answer** Yes, equivalent

WORDS equivalent expressions (*expresiones equivalentes*) • counterexample (*contraejemplo*) • **Distributive Property** (*propiedad distributiva*) • **Commutative Property** (*propiedad conmutativa*)

PRACTICE

show your work • label units

1. Evaluate each expression when $y = 6$. Which two might be equivalent?

a. $4(y + 3) =$ _____

b. $4y + 12 =$ _____

c. $4y + 3 =$ _____

2. **Rental Formulas.** Shop A uses $4b + 7 + 5b$ for the cost of renting b bikes. Shop B uses $9b + 7$.

a. Substitute $b = 2$. A: _____ B: _____

b. Substitute $b = 6$. A: _____ B: _____

c. Are they equivalent? How do you know for every b ? _____

d. Which property lets you reorder $4b + 7 + 5b$? _____

3. **One Test Is Enough.** Compare $5(k + 1)$ and $5k + 1$ using $k = 3$. Are they equivalent?

Hint Do the parentheses first in $5(k + 1)$.

Show your work.

Equivalent? _____

5. **Theo's Claim.** Theo says $7 + 2j$ and $9j$ are equivalent because $7 + 2 = 9$. Test $j = 3$ to justify or correct his claim.

Hint Substitute 3 into both expressions.

Show your work.

Values _____

4. **Properties.** Look at $4 + (m + 9)$ and $(4 + m) + 9$.

Hint Only the grouping changed.

a. Which property is shown? _____

b. Test both with $m = 4$. _____

6. **Two Test Values.** Compare $8v + 4$ and $4(2v + 1)$ using $v = 1$ and then $v = 5$. Are they equivalent?

Hint Matching values support it; a property proves it.

Show your work.

Values _____

7. **Lila's Claim.** Lila says $15n - 6n$ and 9 are equivalent because $15 - 6 = 9$. Test $n = 4$ to justify or correct her claim.

Hint Subtracting coefficients leaves a term that still has n .

Show your work.

Values _____

8. **Three Expressions.** Use $2(k + 5)$, $2k + 10$ and $2k + 5$.

- a. Evaluate all three when $k = 3$. _____
- b. Which two are equivalent? _____
- c. Justify with a property, not a test value. _____

FIND THE ERROR

explain the mistake, then fix it

9. **Priya's Work.** Priya tested $4p + 6$ and $10p$ using $p = 1$. Both gave 10, so she said they are always equivalent. Explain her mistake. Test $p = 3$.

TEST PRACTICE

choose the best answer

10. One matching test value is –

- A** not enough to prove equivalence **B** enough to prove equivalence
- C** a counterexample **D** always wrong

11. $5(a + 2)$ is equivalent to –

- A** $5a + 2$ **B** $5a + 10$ **C** $a + 10$ **D** $7a$

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

41 Q	11 T	36 N	17 S	20 N	25 C	44 A	30 Z	13 T	61 O
#2a	#2b	#3	#4b	#5	#6	#7	#8a		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____